

Assigning Reviewers under Rising Load: A Whittle Index for Online Reviewer Assignment

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Abstract

Journals receive submissions on a rolling basis, and submission rates continue to rise. Unlike conferences, which optimize reviewer assignments over a complete batch of submissions, journals must assign each paper as it arrives. Existing systems therefore assign the best-matched available reviewers, ignoring the opportunity cost of committing reviewers who may be a better match for future papers. This cost grows with load.

We model journal reviewer assignment as an online task assignment problem with reusable resources: each arriving paper must immediately be assigned a fixed number of reviewers from a finite pool, and assigned reviewers remain unavailable until they complete their reviews. Modeling each reviewer as an arm of a restless multi-armed bandit, we prove the problem is indexable and derive a closed-form Whittle index policy. We prove asymptotic optimality under a standard fluid scaling when reviewer scores are independent, and characterize the opposite extreme, a rank-one regime in which the index provably reverses Greedy assignment.

We also collect and release a trace of 6,588 TMLR submissions, with arrival times, review durations, and TF-IDF similarities to a pool of 1,385 reviewers. On this trace the index beats Greedy by a margin that grows with load, reaching 7.6% in the final year against 16.2% for a clairvoyant policy that knows all future arrivals.

1 Introduction

A review is only useful if the paper reaches reviewers with the right expertise. Getting this right has two parts. First, we must estimate how well each reviewer fits each paper. Second, we must decide which reviewers actually receive each paper.

Most work on automated reviewer assignment studies the first part, the estimation of match quality (Charlin and Zemel 2013; Singh et al. 2023). The second part, deciding which reviewers actually receive each paper, splits by venue: conferences collect every submission by a deadline and solve a batch matching over the full score matrix, while journals treat each paper as a query served on its own, one at a time (Zhao and Zhang 2022).

We focus on one such journal, *Transactions on Machine Learning Research* (TMLR). Papers arrive one at a time, and each must be assigned soon after it arrives, before later papers are seen. Two things change when the deadline disappears. First, the decision becomes online, since the score matrix is revealed one row at a time and an assignment must be

committed without knowing which papers will follow. Second, a reviewer who accepts a paper is unavailable until that review is returned, so the reviewers are reusable resources that a policy must schedule over time rather than a fixed supply it partitions once. These two changes interact, and their interaction is what makes the journal problem hard. Committing a strong reviewer now removes that reviewer from the pool for the length of a review, during which a paper that needs them more may arrive.

The natural policy is Greedy assignment, which serves each query with the best available reviewers and ignores this effect entirely. Its consequences are already recognized. Zhao and Zhang observe that under retrieval-based assignment the reviewers of broad expertise rank highly for most papers and so become overloaded, while the papers served later are left with less expert reviewers (Zhao and Zhang 2022). Both effects are the same omission: Greedy never prices the future value of the reviewer it commits. When review load is light a well-matched reviewer is usually free and the omission costs little, whereas as load grows and strong reviewers become scarce relative to demand it can matter a great deal. We treat it instead as a scheduling problem, and price the commitment directly.

Online assignment with reusable resources has been studied in theory (Dickerson et al. 2021; Sumita et al. 2022; Nanda et al. 2020), and these results give competitive-ratio guarantees against adversarial or stochastic arrivals. Their policies rest on a linear program, solved either once as offline preprocessing or once per arrival, whose size grows with the resource pool. The experiments in these papers use tens of resources, whereas our pool has 1,385 reviewers, so we look for a policy that needs no linear program at all.

We model each reviewer as an arm of a restless multi-armed bandit and assign papers by the Whittle index. The index gives each free reviewer one number, its similarity to the current paper minus the expected cost of making it unavailable. It has a closed form and uses no linear program.

Our contributions are as follows.

- We frame online journal reviewer assignment as a restless bandit. To our knowledge this is the first treatment of its scheduling aspect. We derive a closed-form Whittle index for the model and prove that the single-arm problem is indexable.
- We analyze the index at the two extremes of how far

reviewers agree with one another. Under independent scores it is asymptotically optimal in a large pool. Under *rank-one* scores, where every reviewer ranks papers in the same order, it reverses Greedy and sends every paper below a pool-wide cutoff to the *weakest* available reviewers. The TMLR matrix is far from rank one.

- We release a dataset of 6,588 TMLR submissions over four years with TF-IDF similarity scores for 1,385 reviewers and characterize its arrivals, review times, and rank structure. On this trace the closed-form index matches Greedy at low load and beats it by a growing margin as load rises, reaching +7.6% for the deployable index, while matching or exceeding far more expensive lookahead heuristics.

2 Related Work

Reviewer assignment and peer review. Zhao and Zhang survey three decades of automated reviewer assignment and divide it into assembling the reviewer pool, computing a matching degree between each paper and reviewer, and the assignment optimization (Zhao and Zhang 2022). Most effort has gone to the matching degree (Charlin and Zemel 2013; Singh et al. 2023). For conferences, the assignment is a batch matching solved under fairness constraints (Stelmakh, Shah, and Singh 2021; Kobren, Saha, and McCallum 2019) or for strategyproofness when reviewers also author papers (Xu et al. 2019), all of which assume the full score matrix is available before assigning. The retrieval-based work cited above optimizes which reviewers suit a paper considered on its own, and not the order in which papers are served or the time a reviewer spends committed. Dumais and Nielsen first posed reviewer assignment this way (Dumais and Nielsen 1992), and later work retrieves on richer profiles: publication and domain knowledge (Biswas and Hasan 2007), proposal and reviewer topics in a system deployed at NSF (Hettich and Pazzani 2006), and multiple aspects of expertise so assigned reviewers together cover every subtopic (Karimzadehgan, Zhai, and Belford 2008).

Online matching and reusable resources. Online bipartite matching began with Karp et al. (Karp, Vazirani, and Vazirani 1990), and later work handles edge weights (Fahrbach et al. 2022; Li, Yang, and Ren 2023). Closest to us is online matching with reusable resources, where a matched resource returns after a delay, introduced for ridesharing by Dickerson et al. (Dickerson et al. 2021) and extended since (Nanda et al. 2020; Sumita et al. 2022; Shanks, Yu, and Jacobson 2023; Anonymous 2026). These works give competitive ratios against adversarial or stochastic arrivals. However, in all these instances, the proposed algorithms require solving large LPs, feasible with only 10-100 reusable resources (reviewers), and a similar number of arrival types (papers). (Sumita et al. 2022) note the need for more scalable matching policies in ridesharing applications.

Restless bandits and the Whittle index. Whittle relaxes the per-step activation budget of a restless bandit and prices it with a subsidy, which decouples the arms and gives each one an index (Whittle 1988). Weber and Weiss show the resulting index policy is asymptotically optimal as the number of

arms grows, under a fluid-limit stability condition (Weber and Weiss 1990). The index has a closed form in several scheduling and queueing problems, which is also what we obtain here.

3 Model

A journal has a pool of R reviewers, indexed by $r \in [R]$. Time is discrete. At each step $t \in \mathbb{N}$ at most one paper arrives: a paper arrives with probability $\lambda \in (0, 1]$, and with probability $1 - \lambda$ the step is empty. An arriving paper comes with a vector of similarity scores

$$\mathbf{S}^{(t)} = (S_1^{(t)}, \dots, S_R^{(t)}) \in [0, 1]^R,$$

where $S_r^{(t)}$ measures how well reviewer r matches paper t . These vectors are drawn independently across papers from a fixed distribution $\mathcal{F}$ on $[0, 1]^R$, which we treat as known. We write $\mathbf{S} = (S_1, \dots, S_R)$ for a generic draw from $\mathcal{F}$, so S_r denotes a generic draw of reviewer r 's score.

On arrival, the paper must be assigned to K reviewers, chosen immediately and irrevocably from those currently free. Assigning paper t to a set $A_t \subseteq [R]$ of reviewers earns reward $\sum_{r \in A_t} S_r^{(t)}$, and we set $A_t = \emptyset$ on empty steps. Each assigned reviewer is then busy for D steps and free thereafter. The review duration D is a positive integer random variable drawn independently for each assignment. We assume D has bounded support, so $1 \leq D \leq D_{\max}$ almost surely. If fewer than K reviewers are free, the paper waits in a first-in-first-out queue until K are. Write $\bar{D} := \lambda \mathbb{E}[D]$ for the expected number of papers that arrive during one review.

A policy maps the arriving score vector and the set of free reviewers to an assignment, and we seek one maximizing the long-run average reward

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{r \in A_t} S_r^{(t)}.$$

Our problem is a *restless multi-armed bandit* (RMAB) (Whittle 1988). Each reviewer is an arm with state $x \in \{0, 1, \dots, D_{\max}\}$ equal to the number of steps until it is free, so state 0 means available. Activating an arm in state 0 assigns the arriving paper to that reviewer, collects reward $S_r^{(t)}$, and sends the arm to state D . A passive arm counts down toward 0, and the controller activates K arms on each step that has an arrival. One feature is particular to our setting: the reward of an activation depends on the arriving paper's score vector $\mathbf{S}^{(t)}$, an exogenous context shared by all arms and redrawn each step. The joint state has size $(D_{\max} + 1)^R$, so an optimal policy is intractable to compute.

4 The Whittle Index

The *Whittle index* (Whittle 1988) relaxes the requirement that exactly K arms are active at every step to one that holds only on average, and enforces it with a scalar price ℓ charged on each activation. Under this price the arms decouple, and the index of a state is the price at which activating and staying passive are equally valuable. The construction is valid only when the arm is *indexable*: as ℓ rises, the set of states in

which passivity is optimal must grow monotonically. Not every restless bandit is indexable, so we establish it for our arms.

The rest of this section solves a single reviewer's arm in closed form and verifies indexability. Fix a reviewer r and consider its arm under price ℓ . Only this reviewer's own coordinate matters, so we work with S_r , a generic draw of reviewer r 's score. Busy states are always passive, so the only decision arises in the free state, when the reviewer is available and a paper arrives with score s against it. The two choices are to accept, earning $s - \ell$ and becoming busy for D steps, or to decline, earning nothing and staying free.

Definition 1 (Passive set, indexability, Whittle index). *Let $\mathcal{P}_r(\ell) \subseteq [0, 1]$ be the set of scores at which declining is optimal under penalty ℓ . The arm is indexable if $\mathcal{P}_r(\ell)$ grows monotonically with ℓ ,*

$$\mathcal{P}_r(\ell) \subseteq \mathcal{P}_r(\ell') \quad \text{whenever } \ell \leq \ell'.$$

For an indexable arm the Whittle index of score s is the penalty at which the two actions are equally good,

$$W(s, r) := \inf\{\ell \in \mathbb{R} : s \in \mathcal{P}_r(\ell)\}.$$

The following lemma solves the single-arm problem.

Lemma 1 (Single-arm optimality). *Fix a reviewer r and a score level $s \in [0, 1]$, and set*

$$g := \lambda \mathbb{E}[(S_r - s)^+], \quad \ell := s - \mathbb{E}[D]g.$$

Under penalty ℓ the single-arm problem has optimal average reward g , and an optimal policy accepts a paper exactly when its score is at least s . At score exactly s the two actions are equally good.

The two theorems below follow. All proofs are in the supplementary material.

Theorem 1 (Indexability). *Every arm is indexable.*

Theorem 2 (Whittle index). *The Whittle index of assigning reviewer r to a paper whose score against r is s is*

$$W(s, r) = s - \bar{D} \mathbb{E}[(S_r - s)^+], \quad \bar{D} = \lambda \mathbb{E}[D].$$

The index is the match score s minus the expected score the reviewer gives up by being unavailable. The term $\mathbb{E}[(S_r - s)^+]$ is the expected amount by which one future paper would beat the current one for reviewer r , and $\bar{D}$ counts how many papers arrive while the reviewer is busy.

Definition 2 (Whittle policy). *When a paper with score vector s arrives, assign it to the K free reviewers with the largest values of $W(s_r, r)$, breaking ties arbitrarily.*

The relaxation that produces the index is solved exactly by an index policy: for a suitable price ℓ , assigning every free reviewer whose index exceeds ℓ is optimal for the relaxed problem in which the budget of K assignments per paper holds only on average (Whittle 1988; Hawkins 2003). The Whittle policy above applies this rule to the true problem, where exactly K reviewers must be assigned to every paper.

5 Theoretical Guarantees on Performance

The Whittle index prices a reviewer by the opportunity cost $\mathbb{E}[(S_r - s)^+]$, which is the amount by which a future paper would suit reviewer r better than the paper being assigned. This term can separate two free reviewers only when those reviewers rank papers differently, so how much correlation the score matrix carries across reviewers is what governs the behavior of the index. This section analyzes the two extremes of that axis, and then asks what worst-case analysis can say about either.

Independent reviewers

We first state the regime in which the index is provably good. When the scores of distinct reviewers are independent and take finitely many values, our model is an instance of the multiclass restless bandit framework of Verloop (Verloop 2016). Finiteness is not a restriction in practice, since measured similarity scores are binned. Guarantees for index policies are asymptotic in the size of the pool, so we need a scaling in which the pool grows while the load stays fixed. We obtain one by growing the arrival stream at the same rate as the pool.

Definition 3 (Fluid scaling). *For $n \geq 1$, system n contains n copies of each of the R reviewers, so that the pool has nR reviewers. At each timestep n papers arrive, and each is assigned K reviewers. Write $V_n(\pi)$ for the long-run average reward of policy π in system n , and OPT_n for the long-run average reward of the Bellman-optimal policy in system n .*

Assumption 1 (Global attractor). *The fluid dynamics of the pool under the Whittle policy have a global attractor, that is, the trajectories of the fluid limit converge to a single equilibrium point from every starting point.*

Assumption 1 is the condition under which Weber and Weiss prove asymptotic optimality of the Whittle index (Weber and Weiss 1990). It is assumed rather than derived throughout this literature (Weber and Weiss 1990; Verloop 2016), and Weber and Weiss give an example in which it fails.

Theorem 3. *Suppose the scores of distinct reviewers are independent and take finitely many values, and suppose Assumption 1 holds. Then the Whittle policy is fluid optimal, namely*

$$\lim_{n \rightarrow \infty} \frac{\text{OPT}_n - V_n(\text{Whittle})}{n} = 0.$$

The result follows from (Verloop 2016, Corollary 5.7) once the model is written in multiclass RMAB. We give that translation in the supplementary material.

Here OPT_n is the optimum of the Markov decision process whose state records the residual busy time of all nR reviewers, so the comparison is against the best policy that sees each paper only on arrival, and not against a clairvoyant one.

This guarantee holds only for independent reviewer scores. In Section 6.1, we examine scores from a real submission trace. We find that similarity scores scraped from TMLR submission data report a mean correlation of 0.10 between two reviewers', against 0 under independence.

Perfectly correlated reviewers

At the other extreme every reviewer ranks papers in the same order.

Definition 4 (Rank-one instance). *Every reviewer r has a quality factor $b_r \geq 0$ and every arriving paper a quality factor $A \geq 0$, drawn from a distribution supported on $[a_{\min}, a_{\max}]$. The score of a paper with factor A against reviewer r is $S_r = A b_r$.*

Under a rank-one instance the reviewers differ only by a scalar, so every reviewer orders papers identically and the score vectors of any two reviewers are perfectly correlated. Recall from Theorem 2 that the index is $W(s, r) = s - \bar{D} \mathbb{E}[(S_r - s)^+]$, and write

$$g(a) := a - \bar{D} \mathbb{E}[(A - a)^+],$$

the index of a paper of factor a against a reviewer whose factor is 1.

Theorem 4 (Rank-one index). *Consider a rank-one instance and a paper with quality factor a .*

- (i) *The index of this paper against reviewer r is $W(a b_r, r) = b_r g(a)$.*
- (ii) *The function g is continuous and strictly increasing on $[a_{\min}, a_{\max}]$, and $g(a_{\max}) = a_{\max} \geq 0$.*
- (iii) *$g(a_{\min}) < 0$ if and only if $\bar{D}(\mathbb{E}[A] - a_{\min}) > a_{\min}$. When this condition holds, g has a unique zero $\tau \in (a_{\min}, a_{\max})$, and the Whittle policy assigns the paper to the K strongest available reviewers when $a > \tau$, and to the K weakest available reviewers when $a < \tau$.*

Since the factor $g(a)$ is common to every reviewer, the index either preserves the ordering of the free reviewers by b_r , which is the Greedy ordering, or reverses it, and the sign of $g(a)$ alone determines which. We call the second case in part (iii) *anti-greedy*, since the paper is then assigned to the reviewers that match it worst. Observe that the policy reaches this regime only because assignment is mandatory in our model: there is no reject action, so a paper with $a < \tau$ must still be given K reviewers, and the index spends the weakest ones on it to hold the rest in reserve.

Competitive ratio. Theorem 4 describes how the index orders the reviewers, but it does not say how much reward the reversal costs. We measure this by the competitive ratio, the standard measure of worst-case performance in the online matching literature (Dickerson et al. 2021).

Definition 5 (Competitive ratio). *An instance fixes the model parameters of Section 3: the pool size R , the assignments per paper K , the arrival probability λ , the distribution of the score vectors, and the distribution of D . For a policy π feasible on that instance, let $\text{Val}(\pi)$ be its long-run average reward and OPT the reward obtained by the Bellman-optimal policy. Then*

$$\text{CR}(\pi) := \inf \frac{\text{Val}(\pi)}{\text{OPT}},$$

the infimum taken over all instances with $\text{OPT} > 0$.

Theorem 5. $\text{CR}(\text{Whittle}) = 0$.

Theorem 6. $\text{CR}(\text{Greedy}) = 0$.

Both bounds are witnessed by rank-one instances, which we construct in the supplementary material. In each the policy is given the true parameters of the instance, so the failure is a property of the policy and not of estimating the score distribution from data. Since both ratios are zero, the competitive ratio does not separate the index from Greedy.

6 Dataset

We assemble a trace from TMLR’s public submission history and use it to drive the simulation in the Experiments section below. It supplies the four inputs the simulation needs: the stream of arriving papers, the reviewer pool, the similarity matrix S , and the review-time distribution. All data come from public APIs. The supplementary material gives the full construction and reports summary statistics of the dataset.

Papers. We retrieve every TMLR submission through the OpenReview API v2, recording its forum ID, submission time (cdate), title, and abstract. Deduplicated by forum ID, this gives 6,588 papers from March 2022 to April 2026, taken in arrival order as the paper stream.

Reviewers. We take the 2,197 Area Chairs and Senior Area Chairs of the NeurIPS 2025 Program Committee as candidate reviewers and attach a publication profile to each from Semantic Scholar, matching on author name and keeping up to four recent abstracts per reviewer. Candidates with no matching profile are dropped, leaving 1,385 reviewers and 4,374 abstracts (mean 3.2 each). The supplementary material gives the matching procedure and the disambiguation errors it can make.

6.1 Similarity scores

The score $s_{p,r}$ is the TF-IDF cosine similarity between paper p and reviewer r ’s profile, following Charlin & Zemel (Charlin and Zemel 2013). We scrape the abstracts of every submitted paper and every reviewer’s 4 recent papers. We tokenize these abstracts over a shared vocabulary, encode and compute cosine similarities for each paper-reviewer pair. This yields a similarity score matrix, $S \in [0, 1]^{6588 \times 1385}$. We also release a matrix of BM25-based scores (Robertson and Zaragoza 2009) for our paper-reviewer pairs. BM25 improves on TF-IDF by saturating and normalizing each keyword’s contribution to the cosine similarity. The supplementary material provides a detailed description of the data collections and score computation procedures.

The matrix S realizes the model of Section 3 as follows: row p of S is one draw $\mathbf{S}^{(t)}$ of the score vector, and its entry $s_{p,r}$ is the realized value of $S_r^{(t)}$. We write the random variables of the model in upper case and realized scores from the trace in lower case throughout. To confirm the results do not hinge on this choice of similarity metric, the supplementary material repeats the main experiment with BM25 scores in place of TF-IDF.

Rank structure. The reversal of Theorem 4 requires a rank-one score matrix, in which every reviewer ranks the papers in the same order. We measure how far the TMLR matrix is from that case by the correlation between reviewers’ score vectors.

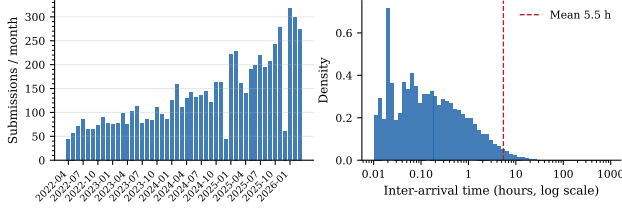


Figure 1: Monthly TMLR submission counts (2022–2026). The rate grows about $3\times$, with an annual holiday pause each December–January.

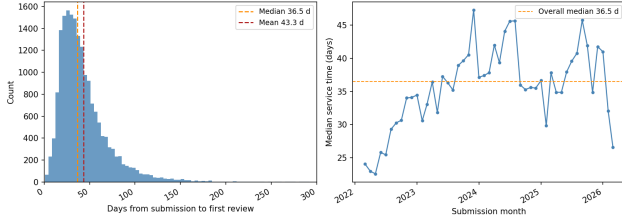


Figure 2: Review times for TMLR (2022–2026). *Left*: distribution of 18,640 submission-to-review durations (median 36.5, mean 43.3 days). *Right*: median service time per submission month.

Averaged over all reviewer pairs, that correlation across the 6,588 papers is 0.10, where rank one would force 1. Fewer than 0.1% of pairs exceed 0.5. The trace therefore sits far from the rank-one regime of Theorem 4.

6.2 Arrival process

Two features of the arrival stream matter for the experiments. First, submissions rates increase over the trace, from 2.2 to 6.4 papers per day (Figure 1). Later years therefore place the pool under heavier load, which is why we report results by year. Second, arrival rates show periodic decreases: we examine weekday variation and holiday effects in the supplementary material.

6.3 Review times

The service time of a review is the time from submission to the posted review. Of the 6,588 submissions, 6,003 have at least one review, giving 18,640 service-time observations with median 36.5 and mean 43.3 days (see Figure 2). 90.1% of decided papers received exactly three reviews. In the simulation, when a paper arrives, each of its assigned reviewers is held for review times drawn i.i.d. from those recorded for that paper in the submission trace.

7 Experiments

We test the index on the TMLR trace and on controlled variants of it that vary load and rank. Throughout, Greedy is the baseline: it is the rule automated systems deploy today. We compare it against two families of alternatives. T-Lookback and T-Lookahead are strong heuristics that solve a bipartite matching over a window of arrivals at every step. We

also present two variants of our Whittle index policy which estimate the opportunity cost in different ways.

7.1 Policies

Every policy assigns each arriving paper p its K reviewers from the free set F_p , immediately and irrevocably, and differs only in how it selects them.

GREEDY. Assign the K free reviewers of highest score, arg top- $K_{r \in F_p} s_{p,r}$.

WHITTLE. Assign the K free reviewers of highest index $W(s_{p,r}, r)$ (Theorem 2). The score distribution $\mathcal{F}$ is not known on the trace, so W is estimated from a window of arrivals, in the two variants below. The exact opportunity-cost term is $\mathbb{E}[D] \mathbb{E}[(S_r - s)^+]$, the expected total excess of the papers arriving during one review, so the windowed sums below are its plug-in estimate under local stationarity.

WHITTLE LOOKBACK (causal). Estimate the opportunity-cost term from the papers that arrived in the preceding review window $[a_p - \mathbb{E}[D], a_p]$:

$$W_{p,r}^{\text{lb}} = s_{p,r} - \sum_{p': a_{p'} \in [a_p - \mathbb{E}[D], a_p]} (s_{p',r} - s_{p,r})^+,$$

and assign the K free reviewers of highest $W_{p,r}^{\text{lb}}$. It uses only papers already arrived, so it is deployable.

WHITTLE ORACLE (non-causal). Identical, but over the following window $(a_p, a_p + \mathbb{E}[D])$:

$$W_{p,r}^{\text{or}} = s_{p,r} - \sum_{p': a_{p'} \in (a_p, a_p + \mathbb{E}[D])} (s_{p',r} - s_{p,r})^+.$$

It uses scores of papers not yet arrived, so it is an upper bound on any causal estimate.

T-LOOKBACK (causal) and T-LOOKAHEAD (non-causal).

Over a window $\mathcal{W}_p$ of T papers, solve the maximum-weight bipartite matching

$$\begin{aligned} \max_{x \in \{0,1\}^{|\mathcal{W}_p| \times R}} \quad & \sum_{i \in \mathcal{W}_p} \sum_{r \in F_p} s_{i,r} x_{i,r} \\ \text{s.t.} \quad & \sum_r x_{i,r} = K \quad \forall i, \quad \sum_i x_{i,r} \leq 1 \quad \forall r, \end{aligned}$$

and assign paper p the reviewers $\{r : x_{p,r} = 1\}$. T-Lookback’s window is the previous T papers; T-Lookahead’s is the next T . Each solves this program at every arrival. We use $T = 50$.

7.2 The TMLR trace

Because the arrival stream is non-stationary (see “Arrival process” above), the score distribution $\mathcal{F}$ is unknown, so we use the windowed estimators WHITTLE LOOKBACK and WHITTLE ORACLE defined above. We process papers in arrival order, draw each assigned reviewer’s review duration D from the empirical review-time distribution (“Review times” above), and assign $K = 3$ reviewers per paper. The durations enter the index only through $\bar{D}$.

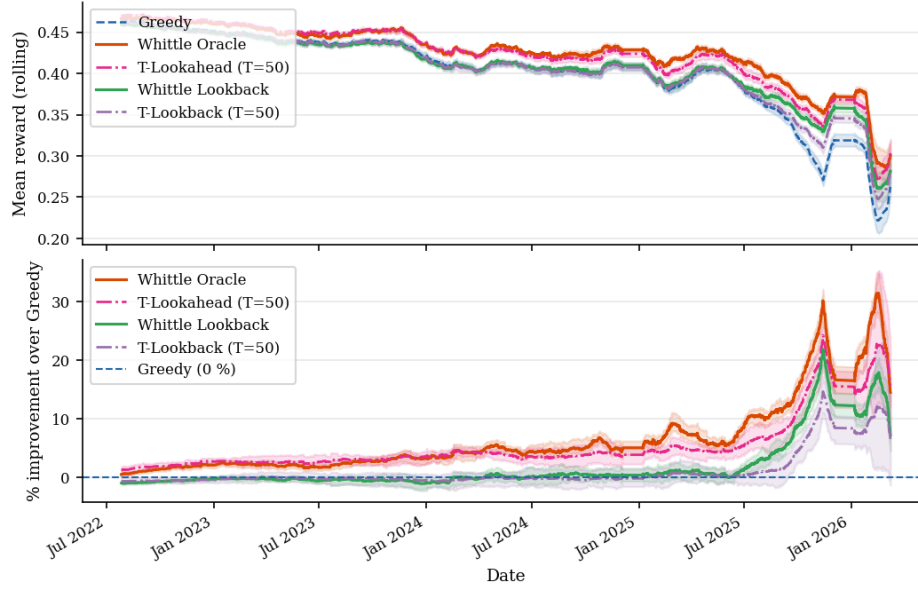


Figure 3: Rolling mean reward on the TMLR trace (window 500 papers, TF-IDF scores, 1,385 reviewers, $T = 50$). Shading is a 95% CI over 20 seeds.

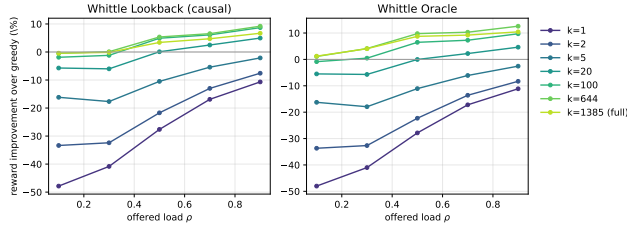


Figure 4: Rank ablation on the real TMLR similarity matrix, truncated by SVD to rank k . Mean over 3 seeds.

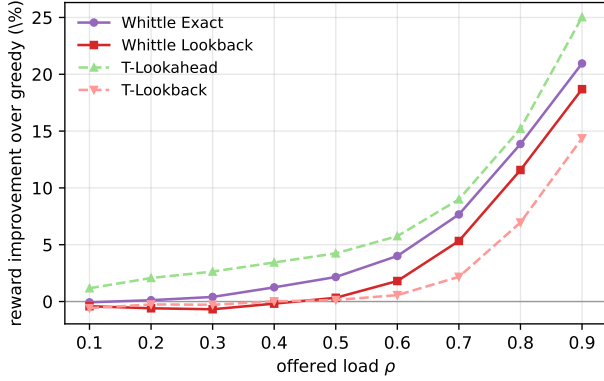


Figure 5: Load ablation under stationary arrivals. $K = 3, 5$ seeds.

Table 1 reports per-year mean reward relative to Greedy; since the arrival rate grows about threefold, later years are

Table 1: Per-year mean reward, percentage improvement over Greedy ($K = 3$). Non-causal policies (W. Oracle, T-Lkah) are upper bounds; causal policies (W. LB, T-LB) are deployable. $T = 50$.

Year (n)	Non-causal		Causal	
	W. Oracle	T-Lkah	W. LB	T-LB
2022 (642)	+1.0%	+1.6%	−0.7%	−0.6%
2023 (1,079)	+2.3%	+2.7%	−0.5%	−0.2%
2024 (1,571)	+5.1%	+4.0%	+0.1%	−0.3%
2025 (2,344)	+11.5%	+8.5%	+5.1%	+2.9%
2026 (952)	+16.2%	+13.8%	+7.6%	+6.0%

higher load. Figure 3 shows the same comparison as a rolling average.

We note three findings. First, the gain grows with load, from near zero in 2022 to +16.2% (oracle) and +7.6% (lookback) in 2026, consistent with the load term of the index and with TMLR being far from rank one (“Rank structure” above). Second, the closed-form index exceeds the window heuristics on both sides of the causal boundary: Whittle Lookback (+7.6%) over T-Lookback (+6.0%), and Whittle Oracle (+16.2%) over T-Lookahead (+13.8%), though the latter solve a bipartite matching at every arrival. Third, the causal policies track Greedy to within 1% through 2024, and both turn clearly positive from 2025.

To check that the improvement does not depend on the particular similarity metric, we recompute the scores with BM25 and repeat the load sweep on the same pool. The improvement follows the same pattern, remaining near zero at low load and growing with load, although its magnitude is smaller than under TF-IDF (supplementary material).

7.3 Load ablation

In Section 7.2 the causal policies fall below Greedy in the early years and rise above it in the later ones. Since the arrival rate grows across those years while the number of papers the index estimates from grows with it, the shortfall could come from either, and the per-year comparison cannot tell them apart.

We replace the arrival stream with one drawn from the model of Section 3, so the arrival rate is constant by construction. Papers arrive as a Poisson process at rate $\lambda = \rho R / (K \mathbb{E}[D])$. Each arriving paper’s score vector is drawn uniformly with replacement from the 6,588 rows of S , so score vectors are independent across papers and $\mathcal{F}$ is fixed and known. Review durations are drawn from the empirical distribution of Section 6.3. We assign $K = 3$ reviewers from the pool of 1,385, queue arrivals in FIFO order, and sweep ρ from 0.1 to 0.9 over five seeds.

Because $\mathcal{F}$ is known, we run the index twice. WHITTLE EXACT computes $\mathbb{E}[(S_r - s)^+]$ over all 6,588 rows. WHITTLE LOOKBACK replaces it with the windowed sum of Section 7.1. The two are otherwise identical, so any difference between them is due to the estimate.

Figure 5 reports mean reward relative to Greedy. Below $\rho = 0.3$ only T-Lookahead separates from Greedy, since a well-matched reviewer is almost always free at that load and the opportunity cost the index prices is near zero. The gains grow with load, and at $\rho = 0.9$ the deployable Whittle Lookback is +18.7% over Greedy. Whittle Exact is above Whittle Lookback at every load, and among the causal policies Whittle Lookback is above T-Lookback once $\rho \geq 0.5$. T-Lookahead, which reads future arrivals, is ahead of all of them.

Since the arrival rate is constant here, and since Whittle Lookback still falls below Greedy at low load whereas Whittle Exact does not, the shortfall is produced by the windowed estimate, which at $\rho = 0.1$ is formed from 46 papers, rather than by the growing arrival rate of the trace.

The score vectors are drawn independently here, so this construction removes the topic drift of the trace along with its arrival non-stationarity and cannot separate the two.

In addition to improving the average match, the index distributes load more evenly across the pool. At the trace’s natural load, it reduces the Gini coefficient of per-reviewer assignment counts from 0.32 under Greedy to 0.21. Whereas Greedy leaves 32 of the 1,385 reviewers unused over the full trace, the index assigns papers to all of them.

7.4 Rank ablation

The anti-greedy failure of Theorem 4 is a property of rank-one similarity. Real scores are higher rank (Section 6.1), so we ask how the failure depends on rank once real arrivals and real service times are held fixed. We take the real TMLR similarity matrix, truncate its SVD to rank k , and rerun the load ablation of Section 7.3 at each k , sweeping the same range of loads. At $k = 1385$, the matrix’s full rank, this reproduces the load ablation’s own numbers, which also checks the two experiments are consistent with each other.

Figure 4 reports Whittle Lookback’s reward improvement over Greedy at each (k, ρ) pair. Two effects combine. First,

low rank is costly at every load: at $k = 1$ the index loses between 10.7% and 47.8% to Greedy, and the loss stays negative through $k = 5$ at every load we test. Second, rank alone does not decide the outcome. At full rank the index performs within 1% of Greedy at low load. It only overtakes it once $\rho \geq 0.5$, matching the threshold found in Section 7.3. Thus, the index needs both enough rank and enough load to beat Greedy. Both these conditions hold in the recent years of our TMLR submission trace.

8 Conclusion

Given the rising submission loads faced by peer review journals, this work models online reviewer assignment as an online scheduling bandit problem. We propose a scalable closed-form Whittle index policy: for each free reviewer, the Whittle index is the match score minus the *opportunity cost* of assigning the reviewer to the current arrival. The index’s improvement over Greedy grows with load, and decreases with correlation of scores across reviewers.

Real peer review has constraints this model does not include. Conflicts of interest remove specific reviewers from specific papers, which our free set does not track. Reviewers may decline assigned papers. Our work does not consider fairness of the assignment to arriving papers.

We release the TMLR trace hoping it supports both directions. On the estimation side, a better real-time approximation of the opportunity-cost term could close the gap between the deployable Lookback and the oracle in Table 1. On the modeling side, extending the index to reviewers who can refuse, to conflicts of interest, or to workload caps needs new indexability arguments beyond the ones given here.

Use of AI systems

Claude Code was used in preparing this paper. It assisted writing of data scraping and experimental simulation code released with the paper. It was also used to edit and polish author-written text. The authors reviewed and revised all text and code and are responsible for the entire content.

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Supplementary Material: Assigning Reviewers under Rising Load: A Whittle Index for Online Reviewer Assignment

Keerthana Gurushankar

1 Proofs for the Whittle index

Throughout this section we fix a reviewer r and write $m := \mathbb{E}[D]$ for the mean review duration, so that $\bar{D} = \lambda m$. Recall from Definition 1 of the main paper that $\mathcal{P}_r(\ell) \subseteq [0, 1]$ denotes the set of scores at which declining is optimal under penalty ℓ , that the arm is indexable when $\mathcal{P}_r(\ell) \subseteq \mathcal{P}_r(\ell')$ for $\ell \leq \ell'$, and that the Whittle index is then $W(s, r) = \inf\{\ell \in \mathbb{R} : s \in \mathcal{P}_r(\ell)\}$.

Lemma 1 (Single-arm optimality). *Fix a reviewer r and a score level $s \in [0, 1]$, and set*

$$g := \lambda \mathbb{E}[(S_r - s)^+], \quad \ell := s - \mathbb{E}[D] g.$$

Under penalty ℓ the single-arm problem has optimal average reward g , and an optimal policy accepts a paper exactly when its score is at least s . At score exactly s the two actions are equally good.

We argue by exhibiting a solution (g, h) of the average-reward optimality equation, with g the average reward per step and h the relative value function. The arm's rewards are bounded, so a solution with h bounded certifies that g is the optimal average reward and that any policy attaining the maximum is optimal (Puterman 1994, Theorem 8.4.1), and we refer to Sennott (Sennott 1999, Chapter 7) for the general state space that the free state requires here.

Proof of Lemma 1. Let $g = \lambda \mathbb{E}[(S_r - s)^+]$ and $\ell = s - mg$ as in the statement. The state of the arm is x when it is busy with x steps remaining. When it is free we distinguish the two events that can occur on a step, writing $(0, \emptyset)$ for a step with no arrival and $(0, s')$ for a step on which a paper of score s' arrives. Write $\bar{h}$ for the expected relative value of a free arm at the start of a step, before it is known whether a paper arrives, so that $\bar{h}$ plays the role of $h(0)$ in the transitions below. Consider the candidate

$$h(x) := -xg \quad (x \geq 1), \quad h(0, \emptyset) := -g, \quad h(0, s') := (s' - s)^+ - g,$$

and normalise by $\bar{h} = 0$. We verify the optimality equation state by state.

Busy states. A busy arm earns nothing and moves one step closer to being free, so for $x \geq 2$ the equation reads $g + h(x) = h(x - 1)$, and indeed $g + h(x) = g - xg = -(x - 1)g = h(x - 1)$. For $x = 1$ the arm is free at the next step, so the equation reads $g + h(1) = \bar{h}$, and indeed $g + h(1) = g - g = 0 = \bar{h}$.

Free state, empty step. No reward is earned and the arm stays free, so the equation reads $g + h(0, \emptyset) = \bar{h} = 0$, which holds by the definition of $h(0, \emptyset)$.

Free state, arrival of score s' . Accepting earns $s' - \ell$ and sends the arm to state D , whose expected relative value is $\mathbb{E}[h(D)] = -g \mathbb{E}[D] = -mg$, since $D \geq 1$ almost surely. Declining earns nothing and leaves the arm free, which is worth $\bar{h} = 0$. Hence the equation reads

$$g + h(0, s') = \max\{(s' - \ell) - mg, 0\}.$$

Since $\ell = s - mg$, the accept branch equals $(s' - \ell) - mg = s' - s$, so the right-hand side is $(s' - s)^+$, which is $h(0, s') + g$ by definition. The equation therefore holds.

Normalisation. It remains to check that $\bar{h} = 0$ is consistent. A step carries an arrival with probability λ and is empty otherwise, so

$$\bar{h} = \lambda \mathbb{E}[h(0, S_r)] + (1 - \lambda) h(0, \emptyset) = \lambda \left(\mathbb{E}[(S_r - s)^+] - g \right) + (1 - \lambda)(-g) = \lambda \mathbb{E}[(S_r - s)^+] - g,$$

which vanishes precisely because $g = \lambda \mathbb{E}[(S_r - s)^+]$.

Since h is bounded, the verification theorem applies and g is the optimal average reward under penalty ℓ . The maximum in the free state is attained by accepting exactly when $s' \geq s$, with the two actions tied at $s' = s$, so the stated policy is optimal and the arm is indifferent at score s . $\square$

Theorem 1 (Indexability). *Every arm is indexable.*

Theorem 2 (Whittle index). *The Whittle index of assigning reviewer r to a paper whose score against r is s is*

$$W(s, r) = s - \bar{D} \mathbb{E}[(S_r - s)^+], \quad \bar{D} = \lambda \mathbb{E}[D].$$

Proof of Theorems 1 and 2. Write $\widetilde{W}(s, r) := s - \bar{D} \mathbb{E}[(S_r - s)^+]$ for the candidate index. Since $\bar{D} = \lambda m$ and $g = \lambda \mathbb{E}[(S_r - s)^+]$, this candidate equals $s - mg$, which is the penalty ℓ of Lemma 1. By that lemma, at penalty $\ell = \widetilde{W}(s, r)$ an optimal policy declines exactly the scores below s and is indifferent at s .

We first record that $\widetilde{W}(\cdot, r)$ is continuous and strictly increasing. Observe that the map $s \mapsto \mathbb{E}[(S_r - s)^+]$ is nonincreasing and 1-Lipschitz, because $|(u - s)^+ - (u - s')^+| \leq |s - s'|$ pointwise. Continuity of $\widetilde{W}(\cdot, r)$ follows at once, and for $s < s'$ we get

$$\widetilde{W}(s', r) - \widetilde{W}(s, r) = (s' - s) + \bar{D} \left(\mathbb{E}[(S_r - s)^+] - \mathbb{E}[(S_r - s')^+] \right) \geq s' - s > 0.$$

Now fix $\ell \in \mathbb{R}$ and consider the passive set $\mathcal{P}_r(\ell)$. If ℓ lies in the range of $\widetilde{W}(\cdot, r)$ on $[0, 1]$, then by strict monotonicity and continuity there is a unique s with $\widetilde{W}(s, r) = \ell$, and Lemma 1 says that declining is optimal exactly on $\{s' : s' < s\}$ together with the indifference point s itself. Monotonicity turns this set into $\{s' : \widetilde{W}(s', r) \leq \ell\}$. If instead ℓ is below the range, no score is declined and the passive set is empty, while if ℓ is above the range every score is declined and the passive set is all of $[0, 1]$. Both cases agree with the same formula, so for every $\ell \in \mathbb{R}$,

$$\mathcal{P}_r(\ell) = \{s' \in [0, 1] : \widetilde{W}(s', r) \leq \ell\}.$$

This description makes $\mathcal{P}_r(\ell) \subseteq \mathcal{P}_r(\ell')$ whenever $\ell \leq \ell'$, and busy states are passive under every penalty, so the arm is indexable and Theorem 1 follows. Finally, by the definition of the index,

$$W(s, r) = \inf \{ \ell : s \in \mathcal{P}_r(\ell) \} = \inf \{ \ell : \widetilde{W}(s, r) \leq \ell \} = \widetilde{W}(s, r),$$

which is Theorem 2. $\square$

2 Proof of fluid optimality

Theorem 3. *Suppose the scores of distinct reviewers are independent and take finitely many values, and suppose the fluid dynamics of the pool under the Whittle policy have a global attractor (Assumption 1 of the main paper). Then the Whittle policy is fluid optimal under the fluid scaling of Definition 3 of the main paper, in which system n contains n copies of each of the R reviewers and receives n independent copies of the arrival stream, namely*

$$\lim_{n \rightarrow \infty} \frac{\text{OPT}_n - V_n(\text{Whittle})}{n} = 0.$$

Proof. We exhibit the model as a multiclass restless bandit with a fixed population in the sense of Verloop (Verloop 2016), and then invoke the asymptotic optimality of Whittle’s index policy for such bandits (Verloop 2016, Corollary 5.7).

Each reviewer is an arm, so the population is fixed at nR arms in system n and no arm ever arrives or departs. Two reviewers belong to the same class when S_r has the same distribution, so there are at most R classes, one per distinct reviewer, and the class of an arm does not change over time.

An arm is in state $x \in \{1, \dots, D_{\max}\}$ when it is busy with x steps of service remaining, and when it is free we record, in addition, the score of the paper offered to it on the current step. Both hypotheses of the theorem enter here. Because the scores of distinct reviewers are independent, the score vector, which is an exogenous context shared by all arms in the general model, factorises into R mutually independent per-arm streams, so that arm r ’s stream carries no information about any other arm. Because the scores take finitely many values, recording the offered score leaves the per-arm state space finite, which is what Verloop’s fixed-population results require. The transitions are then internal to the arm: from the free state the active action assigns the paper, earns the offered score, and sends the arm to state D , while the passive action leaves it free, and a busy arm is passive and moves from x to $x - 1$.

Since Verloop’s results are stated for average-cost minimisation, we assign cost $-s$ to activating an arm offered score s and cost 0 to every passive action, both bounded because $s \in [0, 1]$. The scaling of the theorem multiplies the arrival rate and the pool by a common factor, which is Verloop’s fluid scaling with parameter $r = n$: system n holds n copies of each reviewer and receives $\text{Bin}(n, \lambda)$ papers per step, so that $K \text{Bin}(n, \lambda)$ arms are activated per step, which is $nK\lambda$ in expectation.

It remains to check the hypotheses of (Verloop 2016, Corollary 5.7).

- (i) *Indexability for the average-cost criterion* holds by Theorem 1.
- (ii) *The unichain condition*, which is Assumption 5.5 of (Verloop 2016), holds for every class under every policy. Indeed, a busy arm reaches the free state in at most $D_{\max}$ steps whatever the policy does, and the free state is reachable from every state, so the chain has exactly one recurrent class, the one containing the free state. Verloop’s remaining technical condition, Condition 4.12, is implied by the unichain condition for a fixed population.
- (iii) *The index policies coincide*. Under the cost convention above the average-cost Whittle index of an arm offered score s is $-W(s, r)$, with $W(s, r) = s - \bar{D} \mathbb{E}[(S_r - s)^+]$ as in Theorem 2. Verloop’s index rule activates the arms of *smallest* cost index, which are the arms of largest $W(s, r)$, so Verloop’s Whittle policy is the Whittle policy of the main paper.
- (iv) *The global attractor property*, which is Condition 4.10 of (Verloop 2016), is the global attractor assumption of the theorem.

By (Verloop 2016, Corollary 5.7) the fluid-scaled reward of the Whittle policy converges to the fluid optimum v^* , so $V_n(\text{Whittle})/n \rightarrow v^*$. The same corollary bounds every other policy by v^* in the limit, so $\limsup_n \text{OPT}_n/n \leq v^*$, while $\text{OPT}_n \geq V_n(\text{Whittle})$ by definition of OPT_n gives the matching lower bound. Hence $\text{OPT}_n/n \rightarrow v^*$ as well, and therefore $(\text{OPT}_n - V_n(\text{Whittle}))/n \rightarrow 0$. $\square$

3 Proofs for the rank-one results

A rank-one instance is the opposite extreme to the independent scores of Section 2, since every reviewer then ranks the papers in the same order.

Theorem 4 (Rank-one index). *Consider a rank-one instance (Definition 4 of the main paper), in which every reviewer r has a quality factor $b_r \geq 0$, every arriving paper a quality factor $A \geq 0$ drawn from a distribution supported on $[a_{\min}, a_{\max}]$, and the score of a paper with factor A against reviewer r is $S_r = Ab_r$. Write*

$g(a) := a - \bar{D} \mathbb{E}[(A - a)^+]$ for the index of a paper of factor a against a reviewer whose factor is 1. For a paper with quality factor a :

- (i) The index of this paper against reviewer r is $W(a b_r, r) = b_r g(a)$.
- (ii) The function g is continuous and strictly increasing on $[a_{\min}, a_{\max}]$, and $g(a_{\max}) = a_{\max} \geq 0$.
- (iii) $g(a_{\min}) < 0$ if and only if $\bar{D}(\mathbb{E}[A] - a_{\min}) > a_{\min}$. When this condition holds, g has a unique zero $\tau \in (a_{\min}, a_{\max})$, and the Whittle policy assigns the paper to the K strongest available reviewers when $a > \tau$, and to the K weakest available reviewers when $a < \tau$.

Proof. For part (i), note that $b_r \geq 0$ lets us factor b_r out of the positive part, so that $(S_r - ab_r)^+ = (Ab_r - ab_r)^+ = b_r (A - a)^+$. Substituting into Theorem 2 gives $W(ab_r, r) = ab_r - \bar{D} b_r \mathbb{E}[(A - a)^+] = b_r g(a)$.

For part (ii), the map $a \mapsto \mathbb{E}[(A - a)^+]$ is nonincreasing and 1-Lipschitz, exactly as in the proof of Theorem 2. Hence g is continuous, and for $a < a'$,

$$g(a') - g(a) = (a' - a) + \bar{D}(\mathbb{E}[(A - a)^+] - \mathbb{E}[(A - a')^+]) \geq a' - a > 0,$$

so g is strictly increasing. Since $A \leq a_{\max}$ almost surely we have $\mathbb{E}[(A - a_{\max})^+] = 0$, whence $g(a_{\max}) = a_{\max} \geq 0$.

For part (iii), every value of A is at least $a_{\min}$, so $\mathbb{E}[(A - a_{\min})^+] = \mathbb{E}[A] - a_{\min}$ and therefore $g(a_{\min}) = a_{\min} - \bar{D}(\mathbb{E}[A] - a_{\min})$, which is negative exactly when $\bar{D}(\mathbb{E}[A] - a_{\min}) > a_{\min}$. Suppose that condition holds. Then $\bar{D}(\mathbb{E}[A] - a_{\min}) > a_{\min} \geq 0$ forces $\mathbb{E}[A] > a_{\min}$, hence $a_{\max} > a_{\min} \geq 0$, and so $g(a_{\max}) = a_{\max} > 0$ strictly. Since g is continuous and strictly increasing, is negative at $a_{\min}$ and positive at $a_{\max}$, it has a unique zero τ , and that zero lies in the open interval $(a_{\min}, a_{\max})$.

It remains to identify the assignment. By part (i) the index of the paper against reviewer r is $b_r g(a)$, which as a function of $b_r \geq 0$ is increasing when $g(a) > 0$ and decreasing when $g(a) < 0$. Strict monotonicity of g gives $g(a) > 0$ if and only if $a > \tau$, and $g(a) < 0$ if and only if $a < \tau$. Since the Whittle policy assigns the paper to the K free reviewers of largest index, those reviewers are the ones of largest b_r when $a > \tau$, and the ones of smallest b_r when $a < \tau$. $\square$

Theorem 4 describes how the index orders the reviewers, but it does not say how much reward the reversal costs. We measure that by the competitive ratio of Definition 5 of the main paper, namely $\text{CR}(\pi) := \inf \text{Val}(\pi)/\text{OPT}$, the infimum of the ratio between π 's long-run average reward and the Bellman-optimal reward taken over all instances of the model with $\text{OPT} > 0$. The next two results show that on this measure the index and Greedy are indistinguishable, since both are as bad as worst-case analysis permits.

Theorem 5. $\text{CR}(\text{Whittle}) = 0$.

Proof. Since rewards are nonnegative we have $\text{CR}(\text{Whittle}) \geq 0$, so it suffices to exhibit, for each $\delta > 0$, an instance on which the ratio falls below δ .

Fix $p \in (0, 1)$ and an integer $d > 1/p$. Take $\lambda = 1$, $D \equiv d$, $K = 1$, and $R = 2(d+1)$ reviewers, of which $d+1$ are *strong* with $b_r = 1$ and $d+1$ are *weak* with $b_r = 0$. Let the paper factor $A \in \{\frac{1}{2}, 1\}$ be i.i.d. with $\Pr[A = 1] = p$.

Since $K = 1$ and each assignment lasts d steps, only the assignments made in the previous d steps are still busy, so at most d reviewers are busy at any time and each class of size $d+1$ always has a free member. Consequently no paper ever queues. Every policy earns at most $A^{(t)} b_r \leq A^{(t)}$ at step t , whereas Greedy attains $A^{(t)}$ at every step because a strong reviewer is always free. Greedy is therefore pathwise optimal, and $\text{OPT} = \mathbb{E}[A] = p + \frac{1}{2}(1 - p)$.

Here $\bar{D} = \lambda \mathbb{E}[D] = d$, so by Theorem 4 a paper of factor a has index $b_r g(a)$ with $g(a) = a - d \mathbb{E}[(A - a)^+]$. At $a = \frac{1}{2}$ only the value $A = 1$ contributes to the positive part, so $\mathbb{E}[(A - \frac{1}{2})^+] = \frac{p}{2}$ and

$g(\frac{1}{2}) = \frac{1}{2}(1 - dp)$, which is negative because $dp > 1$. At $a = 1$ the positive part vanishes and $g(1) = 1$. Every weak reviewer has index $0 \cdot g(a) = 0$ for both values of a . A paper with $A = \frac{1}{2}$ is therefore assigned to a weak reviewer, whose index 0 exceeds the strong reviewers' negative index, and it earns 0. A paper with $A = 1$ is assigned to a strong reviewer and earns 1. Hence $\text{Val}(\text{Whittle}) = p$ and

$$\frac{\text{Val}(\text{Whittle})}{\text{OPT}} = \frac{p}{p + \frac{1}{2}(1 - p)} = \frac{2p}{1 + p} < 2p.$$

Given $\delta > 0$, taking $p < \delta/2$ and $d = \lceil 1/p \rceil + 1$ makes this ratio smaller than δ , as required. $\square$

Theorem 6. $\text{CR}(\text{Greedy}) = 0$.

Proof. As before $\text{CR}(\text{Greedy}) \geq 0$, so it suffices to drive the ratio below any $\delta > 0$.

Fix an integer $d \geq 2$ and let $p := 1/d$. Take $\lambda = 1$, $D \equiv d$, any $K \geq 1$, and $R = K(d+2)$ reviewers, split into a *block* of K reviewers with $b_r = 1$ and a *pool* of the remaining $K(d+1)$ with $b_r = 0$. Let $A \in \{\varepsilon, 1\}$ be i.i.d. with $\Pr[A = 1] = p$, where $\varepsilon > 0$. Any policy that assigns at most K pool reviewers per step leaves at least K of them free at every step, since at most Kd of the $K(d+1)$ pool reviewers are busy at any time. Both policies considered below have this property, so again no paper ever queues. Note also that the block's K reviewers are always assigned together and share the deterministic duration d , so they are simultaneously free or simultaneously busy.

Let f be the policy that assigns the whole block whenever the block is free and $A^{(t)} = 1$, and assigns K pool reviewers otherwise. Starting from a free block, each subsequent step is independently a success with probability p , so the wait until the next block assignment is geometric with mean $1/p$, after which the block is busy for d steps. Exactly one block assignment occurs per cycle and it earns K , since each of the K block reviewers earns $1 \cdot 1$, while every pool assignment earns 0. The renewal-reward theorem therefore gives $\text{Val}(f) = K/(1/p + d) = Kp/(1 + pd)$, and hence $\text{OPT} \geq Kp/(1 + pd)$.

Greedy assigns the block whenever it is free, since $A^{(t)} > 0$ for both values of $A^{(t)}$ while every free pool reviewer scores 0. The block is therefore reused as soon as it becomes free, so it cycles with period exactly d and $\text{Val}(\text{Greedy}) = K \mathbb{E}[A]/d$ with $\mathbb{E}[A] = p + (1 - p)\varepsilon$. Combining the two bounds,

$$\frac{\text{Val}(\text{Greedy})}{\text{OPT}} \leq \frac{(p + (1 - p)\varepsilon)(1 + pd)}{pd} \xrightarrow{\varepsilon \rightarrow 0} \frac{1 + pd}{d} = \frac{2}{d},$$

using $p = 1/d$. The left-hand side is strictly increasing in ε and approaches $2/d$ from above as $\varepsilon \rightarrow 0$, so given $\delta > 0$ we take $d = \lceil 2/\delta \rceil + 1$, which makes the limit $2/d$ strictly smaller than δ , and then choose ε small enough that the ratio itself stays below δ . $\square$

4 Dataset construction

Table 1 reports summary statistics of the dataset.

Reviewer profiles. Starting from a list of 2,399 NeurIPS 2025 reviewers, we attach a publication profile to each name as follows:

1. search the Semantic Scholar author database for the name, keeping the candidates that share at least one name token with it;
2. try those candidates in descending order of citation count, at most three of them;
3. accept the first candidate whose recent paper titles contain a term from a fixed 136-term CS/ML keyword list, or whose venue string matches one of 56 ML/AI venue patterns, and keep up to four of that candidate's abstracts.

Table 1: Statistics of the TMLR dataset. Similarity statistics are over all $6,588 \times 1,385$ paper-reviewer pairs.

Description	Value
Papers (total)	6,588
Date range	2022-03-11 – 2026-04-15
Mean arrival rate	4.4 papers/day
Reviewers retained (of 2,418 NeurIPS’25 candidates)	1,385
Reviewer profile abstracts indexed	4,316
<i>TF-IDF similarity</i>	
Mean / std. dev.	0.042 / 0.024
Mean top score per paper	0.189
Median per-reviewer coefficient of variation	0.452
<i>BM25 similarity</i>	
Mean / std. dev.	0.330 / 0.025
Mean top score per paper	0.433
Median per-reviewer coefficient of variation	0.043
<i>Review times (days)</i>	
Papers with ≥ 1 review	6,003
Individual review observations	18,640
Service time: median / mean	36.5 / 43.3

Both lists are given verbatim in the code release. Candidates for which no profile is accepted are dropped, which leaves 1,385 reviewers and 4,316 abstracts, a mean of 3.1 per reviewer. Disambiguation by citation count is imperfect, since for a common name a higher-cited researcher in another field can be selected, which drops the intended reviewer.

TF-IDF similarity. Every submission abstract and every profile abstract is tokenised by lowercasing, splitting on non-alphabetic characters, dropping stopwords and single characters, and applying the Porter stemmer. The corpus over which the term statistics are computed is the union of the 6,588 submission abstracts and the 4,316 profile abstracts, so submissions and profiles share one vocabulary and one set of inverse document frequencies. Term t in document d receives the augmented weight

$$w_{d,t} = \left(0.5 + 0.5 \frac{\text{tf}(t,d)}{\max_{t'} \text{tf}(t',d)}\right) \text{idf}(t), \quad \text{idf}(t) = \log \frac{N_{\text{docs}}}{\text{df}(t)},$$

where $\text{tf}(t, d)$ is the count of t in d , $\text{df}(t)$ the number of documents containing t , and N_{docs} the corpus size. The document vector $\mathbf{v}_d = (w_{d,t})_t$ is then ℓ_2 -normalised. The score of paper p against reviewer r is the best match over r ’s profile abstracts,

$$s_{p,r} = \max_{q \in \text{profile}(r)} \cos(\mathbf{v}_p, \mathbf{v}_q),$$

following the TPMS construction of Charlin and Zemel (Charlin and Zemel 2013). No rescaling is applied: the weights $w_{d,t}$ are nonnegative, so every cosine already lies in $[0, 1]$, and $S \in [0, 1]^{6588 \times 1385}$ as the model requires.

BM25 similarity. For the robustness check of Section 6 we recompute the matrix with Okapi BM25 (Robertson and Zaragoza 2009) over the same tokenisation. Here the corpus is the pool of profile abstracts alone, each abstract being one document, and the submission abstract acts as the query. As above, the score of paper p against reviewer r is the largest BM25 score over r ’s abstracts. BM25 scores are unbounded above, so

the resulting matrix is mapped into $[0, 1]$ by a single global affine rescaling, subtracting the matrix minimum and dividing by the range.

Both matrices cover the same 6,588 papers and the same 1,385 reviewers, so they can be substituted for one another without changing anything else. They rank reviewers differently, however: the median Spearman correlation between the two score vectors of a paper is 0.54, and the two metrics agree on only 10% of the top-three reviewer sets. BM25 is therefore a genuine alternative rather than a monotone relabelling of TF-IDF, which is what makes it a useful robustness check in Section 6.

5 Arrival-process model

The experiments in the main paper use two arrival streams. The trace experiment replays the real submission times, and the load ablation replaces them with a homogeneous Poisson stream at a controlled rate. This section fits a descriptive model to the real stream. It is not used to generate any of the simulations. Its purpose is to substantiate the two claims the main paper makes about the trace, namely that the submission rate grows over the four years and that the stream is not a homogeneous Poisson process, which is what motivates replacing it with a stationary stream when load is the quantity under study.

We model the daily submission count as Poisson with a time-varying rate, fitted by a Poisson GLM,

$$\log \mu_j = \alpha + \beta j + \gamma \mathbf{1}[\text{shutdown}_j] + \eta \mathbf{1}[\text{weekday}_j], \quad (1)$$

where j indexes the 1,498 days of the trace and μ_j is the expected count on day j . We write μ rather than λ here to avoid a clash with the per-step arrival probability of the model. The fitted trend is $e^{365\beta} = 1.51$, so the rate grows by half again each year. The weekday indicator carries $e^\eta = 1.90$. The shutdown indicator marks the annual December–January closure of submissions, which falls inside the trace twice, from 4 December 2024 to 6 January 2025 and from 4 December 2025 to 5 January 2026. Those 67 days contain no submissions at all, so the coefficient γ is not identified and diverges, and we simply record that the fitted rate is zero there. The closure was introduced part-way through the trace, so the two earlier winters are not marked.

To assess fit we use the time-rescaling theorem (Brown et al. 2002): under a correctly specified model the inter-arrival times, rescaled by the fitted compensator, are i.i.d. unit exponentials and so have coefficient of variation equal to 1. Since the model is fitted to daily counts while rescaling needs arrival times, we build the compensator by accumulating the fitted daily rates and interpolating linearly within each day.

Table 2 reports the rescaled coefficient of variation for nested models. The trend-only model *worsens* it, because its compensator accumulates phantom arrivals across the shutdown windows, where the true rate is zero. Adding the shutdown indicator drops the coefficient of variation to 1.23, and adding the weekday term drops it further to 1.16. The full model nonetheless rejects the Poisson null under a Kolmogorov–Smirnov test at $p < 10^{-11}$, and its deviance per degree of freedom remains above 1, so residual clustering survives every term we fit. Figure 1 shows where that clustering sits in rescaled time.

Table 2: Time-rescaling diagnostics. The rescaled coefficient of variation equals 1 under a correctly specified Poisson model.

Model	Rescaled CV	Deviance / df
Raw inter-arrivals (no model)	3.03	—
Trend only	4.06	2.48
Trend + shutdown	1.23	1.87
Trend + shutdown + weekday	1.16	1.58

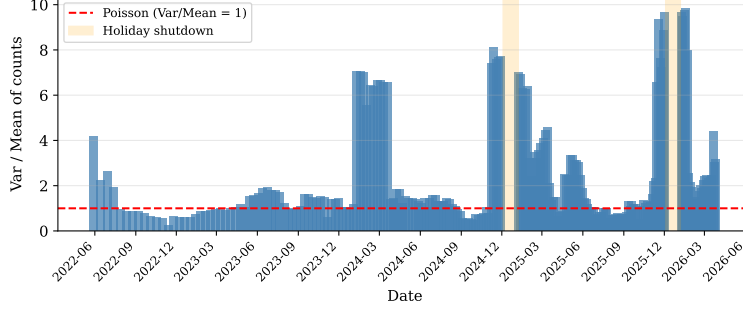


Figure 1: Local index of dispersion of submission counts in compensator-rescaled time, which equals 1 for Poisson counts. Orange bands mark the two shutdown windows.

6 Robustness to the similarity model

The gain reported in the main paper is computed from TF-IDF scores, so we ask whether it survives a different similarity metric. We recompute the score matrix with BM25 as described in Section 4 and rerun the load ablation on it. Because both matrices cover the same 6,588 papers and the same 1,385 reviewers, the comparison needs no matching of pools: the two rows of Table 3 differ in the score matrix and in nothing else. The construction is that of the main paper’s load ablation. Papers arrive as a homogeneous Poisson process at rate $\rho R / (K \mathbb{E}[D])$, each paper’s score vector is drawn uniformly with replacement from the rows of the matrix, review durations are drawn from the empirical distribution, arrivals queue in first-in-first-out order, $K = 3$, and each entry is a mean over five seeds. The arrival stream is a function of the seed and the load alone, so the two metrics are compared on identical streams paper for paper.

Table 3: Reward improvement over Greedy under the two similarity metrics, on the full pool $R = 1,385$ with $K = 3$ and five seeds. Whittle Exact computes $\mathbb{E}[(S_r - s)^+]$ over all 6,588 rows, which the stationary construction makes available, and Whittle Lookback is the deployable windowed estimator of the main paper.

Metric	Policy	$\rho = 0.1$	0.3	0.5	0.7	0.9
TF-IDF	Whittle Exact	−0.1%	+0.4%	+2.2%	+7.7%	+21.0%
	Whittle Lookback	−0.4%	−0.7%	+0.3%	+5.3%	+18.7%
BM25	Whittle Exact	−1.3%	−2.7%	−2.3%	+1.3%	+5.9%
	Whittle Lookback	−1.2%	−2.2%	−1.9%	+1.3%	+5.8%

The load dependence is the same under both metrics: the index sits at or slightly below Greedy while load is low, crosses over, and then gains steadily. What changes with the metric is where the crossover falls and how much the index wins after it. Under TF-IDF the index overtakes Greedy at $\rho = 0.5$ and reaches +18.7% at $\rho = 0.9$, whereas under BM25 it does not overtake Greedy until $\rho = 0.7$ and reaches only +5.8%. The direction of the main paper’s finding therefore survives the change of metric, while its magnitude does not, and we report the weaker number rather than the matched-pool figure.

Table 3 also locates the cause. Under TF-IDF, knowing the score distribution exactly is worth something, since Whittle Exact runs 2.3 percentage points ahead of the windowed estimator at $\rho = 0.9$. Under BM25 the two policies agree to within 0.1 percentage points at every load, so the shortfall is not a failure to estimate the index but an absence of anything to estimate. The two matrices differ sharply in spread: as Table 1 records, the median per-reviewer coefficient of variation is 0.452 under TF-IDF against 0.043 under

BM25. What the index prices is the opportunity cost $\mathbb{E}[(S_r - s)^+]$, which measures how much better a future paper is expected to suit reviewer r than the paper in hand, so it can separate two free reviewers only to the extent that a reviewer’s scores vary across papers. Averaged over reviewers and papers and taken relative to the mean score, that term is 0.252 under TF-IDF against 0.023 under BM25. The correction the index applies to the greedy score is thus an order of magnitude smaller under BM25, and it reorders the free reviewers correspondingly less.

Figure 2 shows the corresponding rolling-reward trace on the real TMLR arrival order.

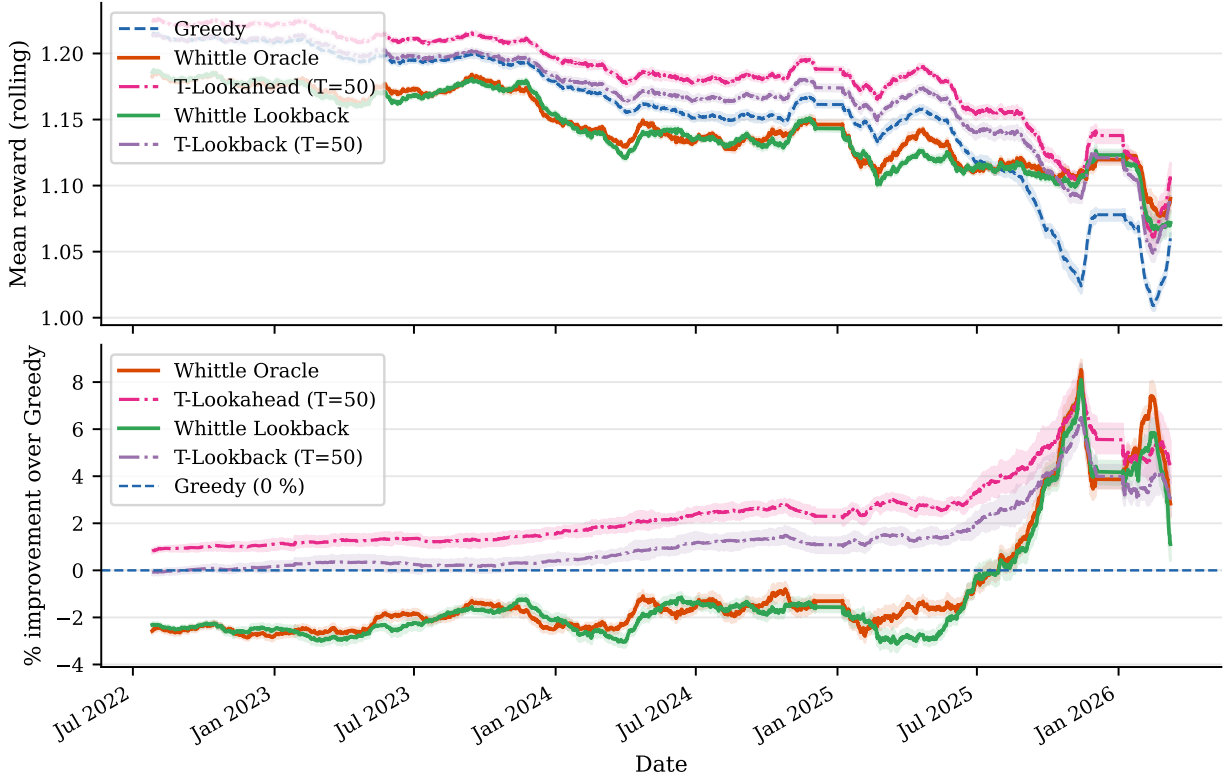


Figure 2: Rolling mean reward on the TMLR trace under BM25 scores, on the same 1,385-reviewer pool as Table 3, with a window of 500 papers and $T = 50$. Shading is a 95% confidence interval over 20 seeds. This is the same construction as the rolling-reward figure of the main paper, which uses TF-IDF scores.

7 Runtime

Whittle Lookback evaluates the index for each free reviewer and takes the top K . T-Lookback and T-Lookahead instead solve a maximum-weight bipartite matching over a window of $T=50$ papers against the free reviewers, once per arrival, using `scipy.optimize.linear_sum_assignment`. We report the measured wall-clock cost of each.

Figure 3 times all four policies on the real TMLR score matrix, subsampling reviewer columns to vary the pool size R . At the pool size used throughout the paper, $R = 1,385$, the amortised cost per arrival is 0.07 ms for Greedy, 0.53 ms for Whittle Lookback, 2.02 ms for T-Lookback and 1.89 ms for T-Lookahead. At $R = 15,000$ the four costs are 0.47 ms, 11.7 ms, 20.0 ms and 19.7 ms respectively.

All experiments in this paper, including this timing comparison, ran on a single machine with a 12th-

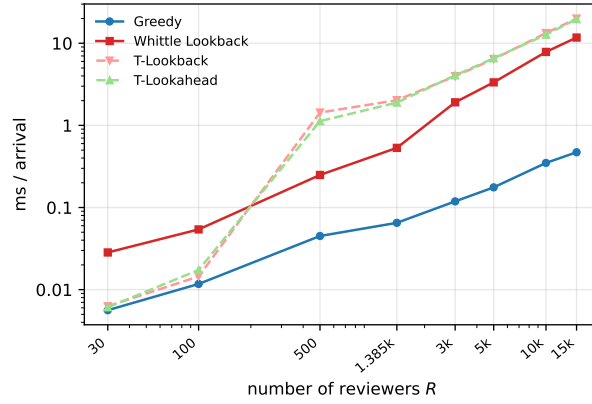


Figure 3: Amortised wall-clock cost per arrival in milliseconds, on a logarithmic scale, for Greedy, Whittle Lookback, T-Lookback and T-Lookahead, against reviewer pool size R , measured on the real TMLR score matrix.

generation Intel Core i7-1265U and 30 GB of memory, using the CPU only, under Ubuntu 24.04 (Linux 6.8) in Python 3.12 with numpy 1.26 and scipy 1.13.